Building and Deploying Sports Data Apps with Docker

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About Me

- Senior at the University of Connecticut 🎓
- Pursuing a Dual Degree in Statistical Data Science and Mechanical Engineering 🔢 ⚙️
- Second year leading a workshop at CSAS.

Outline

1. Why? (5 mins)
2. Containerization vs Virtual Machines (5 mins)
3. Docker (20 mins)
4. Example 1: Visualizing Basketball data with Docker (15 mins)
5. Deploying Apps with Docker (5 mins)
6. Example 2: ML Pipeline (15 mins)
7. Further Reading and Wrap up (5 mins)

Why?

The Problem

Producing reproducable code often has challenges:

- Different OS environments
- Dependency conflicts
- Version mismatches

Especially issues in:

- Data science
- Research
- Team projects

Solution

- Create a completely independant environment develop and deploy program.
- Removes os / version related dependancies

Two ways to implement this:

Containerization and **Virtualization**

Containers vs VMs

Vitrual Machines

- Full **guest OS** per instance
- Heavier (GBs)
- Slower startup
- Strong isolation

Stack:

Hardware
↓
Host OS
↓
Hypervisor
↓
Guest OS
↓
App

Containers

- Share **host OS kernel**
- Lightweight (MBs)
- Fast startup
- Process-level isolation

Stack:

Hardware
↓
Host OS
↓
Docker
↓
Container
↓
App

When to Use

Virtual Machines

- Running **different OS environments**
- Strong **security & isolation**
- **Legacy / monolithic systems**
- OS-level testing and control

Examples:

- Systems requiring strict isolation (e.g., banking systems)
- Testing software on **multiple OS versions** (Windows 10 vs 11)
- Secure environments where workloads must be fully separated

Containers

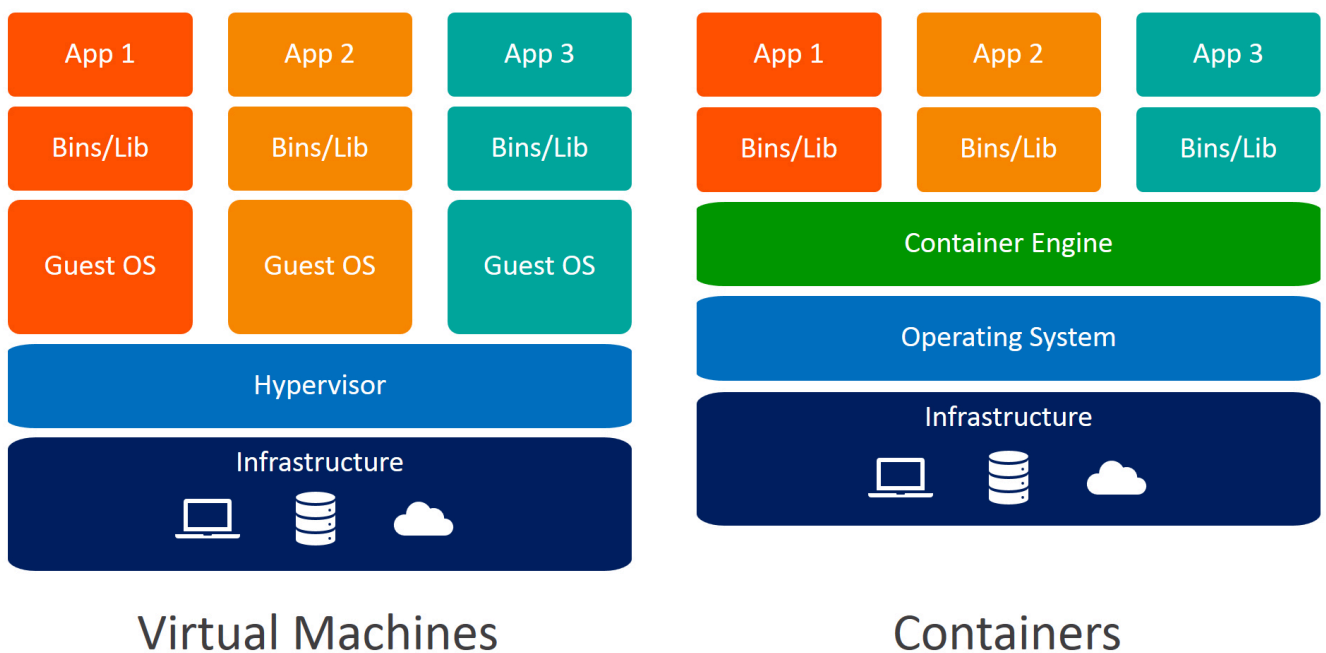
- **Fast development & deployment**
- Microservices architecture
- Reproducibility
- Scalable cloud applications

Examples:

- Running **Jupyter notebooks with fixed dependencies**
- Packaging a **ML model** for deployment
- Spinning up dev environments instantly for team members

Key Takeaway

- **VMs** → full machines, heavy, secure, OS-level control
- **Containers** → lightweight, fast, ideal for modern apps
- Use containers for speed, portability, and development workflows
- Use VMs for strong isolation and different OS needs



Source: BMC Blogs (2026)

Popular Container Platforms

- **Docker** – Build and run containers
- **Kubernetes** – Orchestrate and scale containers
- **Podman** – Secure, daemonless container engine
- **OpenShift** – Enterprise Kubernetes platform
- **Amazon ECS** – AWS container service
- **Singularity (Apptainer)** – Popular in HPC and research environments

Source: Technology Magazine (2024)

Popular Virtual Machine Platforms

- **VMware** – Industry-leading virtualization platform
- **VirtualBox** – Free, open-source VM software
- **Microsoft Hyper-V** – Built into Windows
- **KVM** – Linux-based virtualization (Kernel Virtual Machine)
- **Amazon EC2** – Cloud-based virtual machines on AWS

Source: TechRadar (2024)

Docker

What is Docker

- Platform for **developing, shipping, and running applications**
- First in 2013
- Uses **containers** instead of virtual machines
- Packages:
  - Application code
  - Runtime
  - Libraries & dependencies
- Ensures **consistency across environments** (dev → test → production)

Source: Wikipedia contributors (2026)

Deploying Applications with Docker

- **Bundle everything:** code, dependencies, runtime, configs
- **Run anywhere:** consistent across systems
- **Push to registry:** Docker Hub or private registry
- **Scale easily:** single or multiple containers, works with Kubernetes/Swarm
- **Use cases:** web apps, APIs, databases, ML models

Docker Engine

- Core runtime that **builds and runs containers**
- Responsibilities:
  - Build images
  - Run containers
  - Manage networks & volumes
  - **Available on Linux only.**

Docker Desktop

- User-friendly application for **Windows & macOS**
- Includes:
  - Docker Engine
  - GUI dashboard
  - Kubernetes (optional)
- Handles:
  - Linux VM setup on Mac/Windows need for Docker Engine
  - Networking & file sharing

Steps to Build a Docker Container

1. **Prepare Your Application Code**
   - Organize your code (Python, Node.js, etc.)
   - Include any dependencies file (`requirements.txt`, `package.json`, etc.)
2. **Write a Dockerfile**
   - Setup file that defines the container
3. **Build the Docker Image in Terminal**
   - Docker Image - standardized package that includes all of the files, binaries, libraries, and configurations to run a container.
4. **Start Container from Image in Terminal**

Dockerfile

- A **script** that defines how to build a Docker image
- Contains step-by-step instructions
- Define base image: `FROM python:3.10`
  - Set working directory: `WORKDIR /app`
  - Copy your code into the image: `COPY . .`
  - Install dependencies: `RUN pip install -r requirements.txt`
  - Set default command: `CMD ["python", "app.py"]`

Example:

```
FROM python:3.10
WORKDIR /app
COPY . .
RUN pip install -r requirements.txt
CMD ["python", "app.py"]
```

Dockerfile Commands

Commonly Used Commands:

- `FROM` – base image for your container (`FROM python:3.10`)
- `WORKDIR` – set working directory (`WORKDIR /app`)
- `COPY` – copy files/folders into image (`COPY . .`)
- `RUN` – execute commands during build (`RUN pip install -r requirements.txt`)
- `CMD` – default command when container starts (`CMD ["python","app.py"]`)
- `EXPOSE` – declare network ports (`EXPOSE 80`)
- `ENV` – set environment variables (`ENV PORT=8080`)

Dockerfile Commands

Less Common / Advanced Commands:

- `ADD` – like COPY but can download/extract archives (`ADD file.tar.gz /app/`)
- `ENTRYPOINT` – fixed startup command (`ENTRYPOINT ["python"]`)
- `VOLUME` – mount point for persistent/shared data (`VOLUME ["/data"]`)
- `USER` – switch to non-root user (`USER appuser`)
- `LABEL` – add metadata (`LABEL maintainer="you@example.com"`)
- `ARG` – build-time variable (`ARG VERSION=1.0`)
- `HEALTHCHECK` – container health monitoring (`HEALTHCHECK CMD curl --fail http://localhost:80 || exit 1`)

Building and Running Container

Once files are ready, open terminal/CMD in folder or navigate to folder with files.

1. **Build the Docker Image in Terminal**

```
docker build -t my-app-image .
```

- `build` - builds container image from `Dockerfile`
- `-t` - tags or names the container
- `.` - looks for the `Dockerfile` in the current folder and copies all files.

2. **Start Container from Image in Terminal**

```
docker run -p 5000:5000 my-app-image
```

- `run` - Start the container from the image
- `-p 5000:5000` - Port mapping from `<host-port> : <container-port>`

Output: `localhost:<host-port>` or `http://127.0.0.1:*container-port*/`

Example: http://localhost:5000 or http://127.0.0.1:5000 (Preferred)

Docker Commands

Commonly Used Commands:

- `docker build -t <image_name> .` – build an image from a Dockerfile
- `docker run -p <host>:<container> <image>` – run a container with port mapping
- `docker run --name <container_name> <image_name>` - name and run a container
- `docker ps` – list running containers
- `docker ps -a` – list all containers (running & stopped)
- `docker stop <container_id>` – stop a running container
- `docker rm <container_id>` – remove a container
- `docker images` – list images
- `docker rmi <image_name>` – remove an image

Docker Commands

Less Common / Advanced Commands:

- `docker exec -it <container_id> bash` – open a shell in a running container
- `docker logs <container_id>` – view container logs
- `docker inspect <container_id>` – detailed info about a container or image
- `docker network ls` – list Docker networks
- `docker volume ls` – list Docker volumes
- `docker system prune` – remove unused containers, images, and networks
- `docker-compose up` – start multi-container apps (if using Docker Compose)
- `docker-compose down` – stop and remove multi-container apps

.dockerignore File

- Excludes files/folders from being sent to the Docker build context
- Improves **build speed** and reduces **image size**
- Prevents copying unnecessary or sensitive files

Common Examples: - `.git/`
- `node_modules/`
- `__pycache__/`
- `*.log`
- `.env`

Example:

```
node_modules/
*.log
.env
```

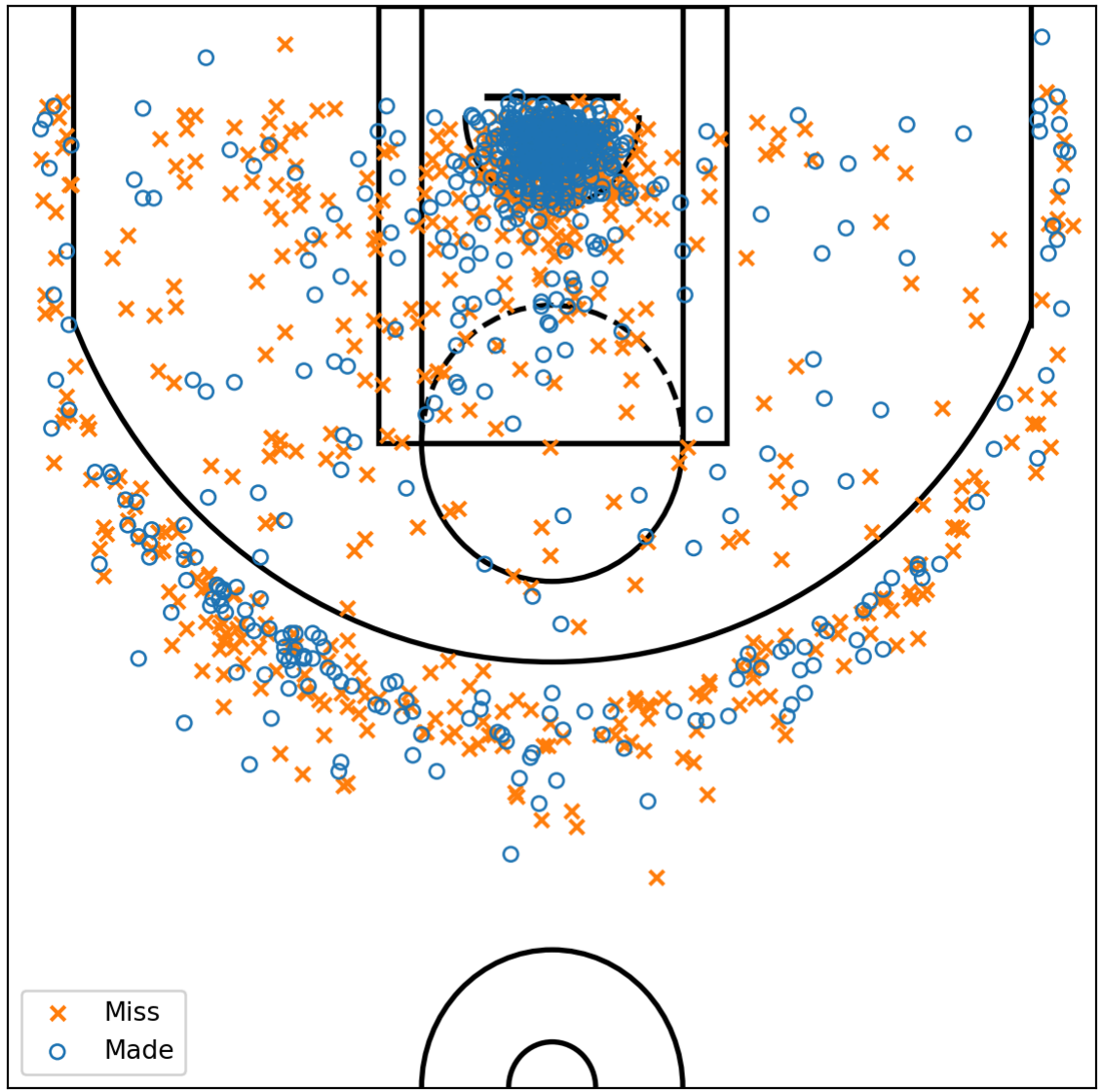
Example 1

Visualizing Basketball data with Docker

Example 1

Create a baseball shot chart in a container.

Lebron James Shot Chart - 2023-24 Season



Step 1: Prepare Your Files

- Python script (e.g., `app.py`)
- Requirements file (`requirements.txt`)
- Dockerfile

All files are available on the github repo for this workshop:

https://github.com/ram200010/CSAS_2026_Docker

Example 1 Folder

app.py

```python
from flask import Flask, jsonify, send_file
from nba_api.stats.endpoints import shotchartdetail
from matplotlib.patches import Circle, Rectangle, Arc
import matplotlib.pyplot as plt


def draw_court(ax=None, color='black', lw=2, outer_lines=False):

    if ax is None:
        ax = plt.gca()

    hoop = Circle((0, 0), radius=7.5, linewidth=lw, color=color, fill=False)

    backboard = Rectangle((-30, -7.5), 60, -1, linewidth=lw, color=color)

    outer_box = Rectangle((-80, -47.5), 160, 190, linewidth=lw, color=color,fill=False)

    inner_box = Rectangle((-60, -47.5), 120, 190, linewidth=lw, color=color,fill=False)

    # Create free throw top arc
    top_free_throw = Arc((0, 142.5), 120, 120, theta1=0, theta2=180,
                         linewidth=lw, color=color, fill=False)
    # Create free throw bottom arc
    bottom_free_throw = Arc((0, 142.5), 120, 120, theta1=180, theta2=0,
                            linewidth=lw, color=color, linestyle='dashed')
```

(Source: Tjortjoglou (2024))

requirments.txt

File that lists the required packages/dependencies.

```
nba_api
matplotlib
flask
```

If you want to specify package versions:

```
nba_api==1.4.1
matplotlib==3.8.0
flask==3.0.0
```

Dockerfile

```
FROM python:3.11

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

EXPOSE 5000

CMD ["python", "app.py"]
```

Build and Run Container

1. Build container

```
docker build -t lebron-shot .
```

2. Run Container

```
docker run -p 5000:5000 lebron-shot
```

Output can be accessed at http://localhost:5000 or http://127.0.0.1:5000

Deploying Apps with Docker

Deploying Apps with Docker

- Share your app with others
- Run on different machines easily
- Deploy to cloud or production

Key Idea:
Build once → run anywhere

Anyone can access your program through **Docker Hub** once published

Deploying Apps with Docker

Once the container image is built:

1. Tag your container

```
docker tag my-app username/my-app:latest
```

2. Login to Docker Hub

```
docker login
```

3. Push the image

```
docker push username/my-app:latest
```

Example Deployed

Simple NBA Player Shot Chart Maker - Web App

Availabe at `ram20010/shot-chart`

Can be run through Docker Hub in Docker Desktop.

or

Can be run directly by running the following.

General Format:

```
docker run -p host-port:container-port username/app-name
```

This Example:

```
docker run -p 5000:5000 ram20010/shot-chart
```

Example 2

ML model in a Container

Example 2

Run Random Forest Classifier model to predict a miss or a shot in a container.

User can input player name and season to predict shots for.

Model will be trained on data from five previous seasons.

Prepare Files

- `main.py` - Gets User Input and executes commands
- `get_data.py` - Contains Function to get train and test datasets
- `model.py` - Runs models and returns classification metrics
- `requirements.txt`
- `Dockerfile`

All files are available on the github repo for this workshop:

https://github.com/ram200010/CSAS_2026_Docker

Example 2 Folder

main.py

```python

from nba_api.stats.static import players
from get_data import get_data
from model import train_rf_shot_model


player_name = input('Enter NBA Player Name: ')
player_season = input('Enter Season to be Predicted: ')

player_info = players.find_players_by_full_name(player_name)
player_id = player_info[0]['id']
player_name = player_info[0]['full_name']

print('Gathering data for ',player_name)

train_data, test_data = get_data(player_id,player_season)

print(train_data.head())

print('Training Random Forest Model')

train_rf_shot_model(train_data,test_data)
```

get_data.py

```python
from nba_api.stats.endpoints import shotchartdetail
import pandas as pd

def get_data(player_id,target_season):

    features = ["PERIOD", "MINUTES_REMAINING","SECONDS_REMAINING","ACTION_TYPE",
    "SHOT_TYPE","SHOT_ZONE_BASIC","SHOT_ZONE_AREA","SHOT_ZONE_RANGE","SHOT_DISTANCE",
    "LOC_X", "LOC_Y", "HTM", "VTM", "TEAM_NAME","SHOT_MADE_FLAG"]

    end_year = int(target_season[:4])

    seasons = [f"{y}-{str(y+1)[-2:]}" for y in range(end_year - 5, end_year)]

    all_data = []

    for season in seasons:

        print(f"Fetching {season}...")

        df = shotchartdetail.ShotChartDetail(
                    team_id=0,
                    player_id=player_id,
                    season_type_all_star="Regular Season",
                    season_nullable=season,
                    context_measure_simple='FGA',
```

model.py

```python
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

def train_rf_shot_model(train_df,test_df):

    features = [
        "PERIOD",
        "MINUTES_REMAINING",
        "SECONDS_REMAINING",
        "ACTION_TYPE",
        "SHOT_TYPE",
        "SHOT_ZONE_BASIC",
        "SHOT_ZONE_AREA",
        "SHOT_ZONE_RANGE",
        "SHOT_DISTANCE",
        "LOC_X",
        "LOC_Y",
        "HTM",
        "VTM",
        "TEAM_NAME"]

    X_train = train_df[features]
```

requirements.txt

```
nba_api
pandas
scikit-learn
```

Dockerfile

```
FROM python:3.11

WORKDIR /app

COPY . .

RUN pip install -r requirements.txt

CMD ["python", "main.py"]
```

Build and Run Interactive Container

1. Build container

```
docker build -t shot-predict .
```

2. Run Container

```
docker run -it shot-predict
```

Further Reading

- Docker: Get Started - Docker Docs
- Docker Tutorial
- What is Docker? - Geeks for Geeks

Questions?

Thank you!

References

BMC Blogs. 2026. “Virtual Machines (VMs) Vs Containers: What’s the Difference?” 2026. https://www.bmc.com/blogs/containers-vs-virtual-machines/.

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